The asymptotic complexity of the mirror reconciliation protocol

Generated from `results/mirror-complexity.md`.

A first-principles derivation of the per-side computation, bandwidth, and latency of `rumors`' mirror protocol, written for a reader who knows basic probability and big-O but has never seen this codebase. Every modeling assumption is pinned to a code location; every approximation is named where it is used. All file references are relative to the repository root (this worktree); line numbers are as of commit `a8a3b7a`.

Headline results (derived in §6, constants from the code):

Axis	Result (expected, uniform leaves)	Empirical fit it refines
Per-side computation	$\Theta\big(\min(256 \cdot D,\, N) \cdot (1 + \log_{256}(N/D))\big)$ child-hash comparisons; equivalently $\Theta(D \cdot (1 + \log(N/D)))$ disputed nodes visited, $\times 256$ sibling fan-out each	$\sqrt{C \cdot D}$: a true upper bound (§7), loose by up to $\sqrt{C/D}/\ln(C/D)$
Bandwidth	$\Theta(D)$ message bodies + the same $\min(256 \cdot D,\, N) \cdot (1 + \log_{256}(N/D))$ hash entries (∼36–64 B each) + ≤ 36 frames of framing	$\sim D$: correct up to the slowly-varying log factor
Latency	$\frac{1}{2} \cdot \log_{256}(2 \cdot D \cdot N) + O(1)$ round trips, hard ceiling ≈ 18 RTT (36 frames), floor 1 RTT after the preamble for identical versions	$\log_{256}(C + D)$: same at $D \approx N$, ∼2× high for $D \ll N$ (both tiny)

where $N = C + D$. One meta-finding up front, developed in §7: at every realistic scale ($N \leq 10^7$), the *exact* level-by-level sum tracks $\sqrt{C \cdot D}$ to within a factor of ∼2–4 across almost the whole (C, D) plane. The smooth $D \cdot \log(N/D)$ form is the asymptote; the discrete base-256 staircase underneath it locally mimics a square root. The empirical fit was not an accident, and wall-clock benchmarks at these scales genuinely cannot refute it — §8 gives measurements that can.

> **Source-of-data flag.** `benches/hash_recon.rs` is *not* a reconciliation benchmark: it measures worst-case root-hash *reconstruction* (32 iterated 8 KiB BLAKE3 wraps vs. 34-byte commitments), motivating the per-node hash memo. The reconciliation measurements live in `benches/gossip_grid.rs` and `benches/in_memory.rs` over the divergence grid in `benches/support/grid.rs`. Everything below assumes the $\sqrt{C \cdot D}$ fit came from that grid.

1 The system, in one page

Each replica holds a **sparse Merkle radix trie**: branching factor 256, fixed depth 32, one path byte per level (`src/tree.rs:18-26`, `src/tree/typed/height.rs:110-117`). A leaf's 32-byte path is $\text{blake3}\big(\text{blake3}(\textit{version}) \,\|\, \text{blake3}(\textit{value})\big)$ — the *version is folded into the path* (`src/tree/typed/path.rs:32-44`), so every insert, even of identical content, lands at a fresh, computationally-uniform position. Single-child runs are path-compressed into a prefix byte string on the node (`src/tree/typed/untyped.rs:25-47`, invariant "branches have ≥ 2 children" at `untyped.rs:78-81, 128-150`). Each node lazily memoizes its subtree hash (`untyped.rs:266-288`) and its version ceiling/floor (`untyped.rs:298-360`); the memos survive copy-on-write clones (`untyped.rs:100-123`). Hashing is compression-invariant: a compressed prefix byte wraps the child hash exactly as a materialized one-child branch would (`src/tree/typed/hash.rs:54-79`, `untyped.rs:268-287`). Leaves hash to a constant (`hash.rs:45-51`) — all leaf-distinguishing content lives in the *path*.

The **mirror protocol** reconciles two replicas' trees over a wire. After a 25-byte raw preamble (`src/tree/traverse/mirror/remote.rs:11-17`) and a framed version handshake, equal versions terminate immediately (`src/tree/traverse/mirror/local.rs:230-241, 268-281`; `remote.rs:279-285`). Otherwise both sides run a synchronized descent. Each side keeps a *zipper* of per-height levels (`src/tree/typed/levels.rs`); each message a side sends descends its own zipper by **two heights** (`local.rs:395-415`; `partition.rs:354-356`), and because the two sides alternate strictly, the conversation's frontier advances **one depth per message, two per round trip**.

Each steady-state message carries three channels (`src/tree/traverse/mirror/message.rs:50-70`, asymmetry matrix at `local.rs:46-58`):

Channel	Content	Receiver's action
`uncertain`	(*prefix*, *hash*) of the sender's disputed-frontier subtrees	merge-join against own children: equal hash $\Rightarrow$ drop; differ $\Rightarrow$ explode grandchildren (recurse); we-lack $\Rightarrow$ `request`; they-lack $\Rightarrow$ `provide`
`requested`	prefixes the sender lacks entirely	answer next message by exploding that node's children into `providing` (`partition.rs:91-142`)
`providing`	whole subtrees, full structure on the wire (`message.rs:50-58`)	insert at the named prefix (`partition.rs:57-85`)

All per-round set operations are single linear merges over sorted flat vectors (`partition.rs:100-115, 199-203`; `src/tree/typed/levels/level.rs`), so per-round work is linear in the frontier and message sizes — no hidden log factors beyond the `imbl OrdMap` child maps (an $O(\log 256)$ constant per child operation, absorbed below).

Termination is in-band: a side is done when its outgoing `requested` and `uncertain` are both empty (`partition.rs:393-410`; `remote.rs:34-43, 364-374, 398-409`); a sender that just emitted an empty-empty message does not even await a reply (`remote.rs:398-409`). The schedule has a hard ceiling — `mirror_connected` (`src/tree/traverse/mirror.rs:126-137`) is `initiator` + `responder` + `open_initiator` + a `seq!` loop of **14 exchange pairs** +

`close_initiator` + `complete_responder` + `complete_initiator`. Counting frames: 2 handshake + `Initiate` (depth 0) + `Opening` (depth 1) + `Exchange` (depth 2) + 28 exchanges (depths 3–30) + one more responder exchange (depth 31) + `Closing` + `Complete` = **36 frames maximum**, covering all 32 byte-levels (the chain lengths are pinned at the type level in `src/tree/traverse/mirror/protocol.rs:477-488`).

Deletions leave no tombstones. Every subtree shipped through `providing` is first filtered by `Unknown::unknown` (`src/tree/traverse/unknown.rs`): a subtree whose version *ceiling* is $\leq$ the counterparty's version vector is something they have already seen — its absence on their side means they deleted it — so it is dropped in $O(1)$ (one memoized-ceiling comparison, `unknown.rs:28-34`), *and the local copy is dropped too* (the Left arm keeps a node only if it survives the filter, `partition.rs:232-239`). A floor check keeps wholly-unknown subtrees intact without rebuilding them (`unknown.rs:38-52`).

2 Model and assumptions

- **(A1) Uniform leaf placement.** Leaf paths are BLAKE3 outputs over (*version*, *value*) (`path.rs:32-44`), and versions are unique per action (`src/tree.rs:299-318`). In the random-oracle model the N leaf paths are i.i.d. uniform 32-byte strings, *regardless of insertion order, batching, or payload skew* — an adversary would need structured BLAKE3 preimages to cluster them. This is the load-bearing probabilistic assumption; see §9.4.
- **(A2) Parameters.** Branching $b = 256$, depth 32 (`tree.rs:21`, `height.rs:110-117`). Hash size $h = 32$ bytes (`hash.rs:17`).
- **(A3) Warm memos.** Subtree hashes and version bounds are memoized reads (`untyped.rs:266-288`); we count a hash *comparison* as $O(1)$. First-touch costs after local mutation are a separate, additive term (§9.1) — the benches warm them in untimed setup (`benches/gossip_grid.rs:119-160`, `src/tree.rs:182-196`).
- **(A4) Dispute drives descent.** Only the *Both-and-hashes-differ* cell of the asymmetry matrix recurses (`partition.rs:243-254`). The Left cell ships a whole subtree in the same message; the Right cell is requested and answered whole one message later (`partition.rs:91-142`). So one-sided differences cost $O(1)$ additional messages once they separate from two-sided content.
- **(A5) Two heights per send, one depth per message.** Verified at `local.rs:395-415` and the alternation in `mirror.rs:126-137`.
- **(A6) No level skipping.** Path compression does **not** shorten the descent: exploding a compressed node pops exactly one prefix byte and yields a singleton child map (`untyped.rs:156-167`). Compression saves hashing and memory, never rounds.
- **(A7) No hash collisions.** A false `uncertain` match (probability $\leq \mathit{pairs} \cdot 2^{-256}$) would silently prune; ignored.
- **(A8) Wire sizes.** An `uncertain` entry at depth k is k raw prefix bytes + 32 hash bytes (`prefix.rs:127-156`, `message.rs:61-63`); a `providing` node is its full borsh structure: per leaf, a version, the payload, and $\leq$ 32 compressed-prefix bytes (`message.rs:21-58`, `untyped.rs:401-438`). Frames carry a 4-byte length prefix (`remote.rs:26-33`).

3 Notation

Symbol	Meaning
b	branching factor, 256
C	live messages present on **both** sides
D	messages present on **exactly one** side, summed over both sides ($D = D_L + D_R$); redacted-on-one-side leaves count (§9.3)
N	$C + D$, the size of the union tree
k	depth = prefix length in bytes, $0 \le k \le 32$; height $H = 32 - k$
Δ_k	number of **disputed** depth-k prefixes: prefixes containing ≥ 1 differing leaf *and* ≥ 1 leaf on the counterparty's side
f_k	expected occupied-children count of a depth-k node, $\approx \min(b, \max(1, N/b^k))$
j_D, j_N	$\log_b D$, $\log_b N$ (depths where b^k crosses D, N)
k^*	separation depth: deepest disputed prefix
W	per-side work / total `uncertain` entries: $\sum_k \Delta_k \cdot f_k$
$\bar{s}$, s_v	mean payload size; serialized `Version` size (not $O(1)$, §9.5)

Logs are base $b = 256$ unless written ln. One decimal decade is $\log_{256} 10 \approx 0.415$ levels; one level spans 2.4 decades. Keep that coarse quantization in mind — it is the source of the staircase in §7.

4 Warm-up: $D = 1,\ N = 10^6$

One side holds a single extra leaf; the trees otherwise agree on 10^6 messages. Trace the protocol:

1. **Handshake** (2 frames): versions differ, descend.
2. **`Initiate`** (depth 0): one root hash. Differs.
3. **`Opening`** (depth 1): the responder *unconditionally* explodes its root and lists all its children (`local.rs:333-353`) — at $N = 10^6$, all 256 slots are occupied ($\approx$ 3,906 leaves each): **256 entries**.
4. The initiator merge-joins: 255 child hashes match and drop; the one slot containing the extra leaf differs (Both-case), so its 256 grandchildren are exploded and listed: **256 entries** at depth 2.
5. The responder finds one depth-2 hash differing; that node has $\approx 10^6/256^2 \approx 15$ occupied children: $\approx$ **15 entries** at depth 3.
6. At depth 3 the extra leaf sits alone in its slot with probability $\approx 1 - N/256^3 \approx 0.94$: a **Left case**. The whole leaf (a path-compressed spine node carrying its remaining ≈ 29 path bytes, version, and payload) ships in `providing`; nothing is exploded; the outgoing `uncertain` and `requested` are empty $\Rightarrow$ **Done**, in-band.

Totals: $\approx 256+256+15 \approx 527$ hash entries on the wire and compared ($\sim$18 KB), 7–8 frames $\approx$ 4 round trips including the preamble, one message body. Three things to notice, which the general analysis formalizes:

- The *visited disputed nodes* number only ≈ 3 (one per level until separation) — $D \cdot (1 + \log_b(N/D))$ with $D = 1$.
- But each visited node costs its **full sibling fan** (≤ 256 entries): the per-node constant is $f_k \approx \min(b, N/b^k)$. This factor-256 is the price of the radix-256 tradeoff: few rounds, fat rounds.
- Separation happened at depth $\approx \log_b(D \cdot N) = \log_b 10^6 \approx 2.5$ — not $\log_b N$ levels of *messages saved*, because termination is the in-band emptiness predicate, not bottoming out at depth 32.

5 Two lemmas

Lemma 1 (prefix occupancy / sharing)**.** *Throw m uniform leaves into the b^k bins at depth k. The expected number of occupied bins is $b^k \cdot (1 - (1 - b^{-k})^m)$, which is $\Theta(\min(b^k, m))$ — at least $(1 - 1/e) \cdot \min(b^k, m)$, at most $\min(b^k, m)$.*

Proof sketch. Linearity of expectation over bins; $(1 - 1/x)^x \leq 1/e$. Applied to the D differing leaves, this bounds how many distinct ancestors they have at each depth: their root-to-leaf paths share prefixes exactly while $b^k < D$, and are essentially disjoint below. Summing, the union of the D differing root-to-separation paths has $\sum_k \min(b^k, D) = \Theta(D \cdot (1 + \log_b(N/D)))$ distinct nodes (the geometric head contributes $\Theta(D)$; the plateau contributes D per level from j_D to the dispute cutoff of Lemma 2). □

Lemma 2 (dispute persistence)**.** *A depth-k prefix is disputed — i.e. recursed into rather than resolved — iff it contains a differing leaf* and *the counterparty owns at least one leaf under it (A4: otherwise it is a Left/Right case and resolves in $O(1)$ messages). Under A1,*

$$\mathbb{E}[\Delta_k] \;\leq\; \min\big(b^k,\; D,\; 2 \cdot D \cdot N \cdot b^{-k}\big)$$

and $\Pr($*any dispute survives depth k*$) \leq 2 \cdot D \cdot N \cdot b^{-k}$.

Proof sketch. The first two bounds are Lemma 1 applied to the differing leaves. For the third: a dispute at depth k requires some differing leaf d and some leaf x on the side opposite d's holder to share a k-byte prefix. Each such (ordered) pair shares it with probability b^{-k} (uniform, independent paths), and there are at most $D_L \cdot (C + D_R) + D_R \cdot (C + D_L) \leq 2 \cdot D \cdot N$ pairs. Union bound / linearity. □

Corollary (separation depth). *$k^* \leq \log_b(2 \cdot D \cdot N) + t$ with probability $\geq 1 - b^{-t}$; $\mathbb{E}[k^*] \leq \log_b(2DN) + \frac{b}{b-1} \cdot \frac{1}{\ln b} \approx \log_b(2DN) + 1$. The fixed 32-level ceiling is unreachable probabilistically: hitting it would need $D \cdot N \approx 256^{30}$. (It exists because paths are 32 bytes; even a full-depth descent is* correct *— equal full paths imply equal (version, value) by construction, `tree.rs:22–26`.)*

Lemma 3 (fan-out)**.** *A depth-k node in the union tree of N uniform leaves has expected occupied-children count $f_k = \Theta(\min(b, \max(1, N/b^k)))$, and the* total *children over any set of Δ depth-k nodes is $\leq \Delta \cdot b$ always and $\Theta(\Delta \cdot f_k)$ in expectation (children counts across disjoint subtrees are negatively associated; we only ever need linearity).* Caution (Jensen): $\mathbb{E}[\min(X, Y)] \leq \min(\mathbb{E}X, \mathbb{E}Y)$*, so products of the* min *envelopes used below are upper envelopes of expectations; the matching lower bounds hold within the $(1 - 1/e)$ constants of Lemma 1.*

6 The level-by-level sum

Per round at depth k, the side processing the counterparty's `uncertain` does (all single-pass, sorted merges — `partition.rs:159-304`):

- drain its frontier once: $O(|\mathit{frontier}_k|)$;
- for each disputed parent, merge-join its own occupied children against the listed hashes: $O(\text{own fan} + \text{listed entries})$, each comparison a memoized-hash `memcmp` (A3);
- explode each hash-differing child's grandchildren into the next level and emit them as the next `uncertain` (`partition.rs:243-254, 364-373`).

So the entries *listed at depth k* are exactly the occupied children of the disputed depth-$(k-1)$ parents, and each side's per-session work, the total `uncertain` traffic, and the zipper/`collapse` churn (`levels.rs:148-197`, linear in total level entries) are all Θ of the same master sum:

$$W \;=\; \sum_{k=0}^{32} \Delta_k \cdot f_k \;\approx\; \sum_k \min\bigl(b^k,\, D,\, 2DN \cdot b^{-k}\bigr) \cdot \min\bigl(b,\, \max(1,\, N/b^k)\bigr)$$

Evaluate by regimes (write $j_D = \log_b D$, $j_N = \log_b N$; assume first $D \le N/b^2$ so all three regimes are non-empty):

Regime	Depths	Term	Contribution
Head (paths shared)	$k \le j_D$	$b^k \cdot b$	geometric, $\approx b \cdot D \cdot (1 + 1/b + \dots) \approx b \cdot D$
Plateau (paths distinct, fans saturated)	$j_D < k \le j_N - 1$	$D \cdot b$	$b \cdot D \cdot (\log_b(N/D) - 1)$
Tail (fans thin out, disputes die)	$k > j_N - 1$	$D \cdot N \cdot b^{-k} \cdot O(1)$	geometric, $\approx \Theta(D \cdot b)$ at the shoulder, then $\div 256$ per level

Theorem 1 (per-side computation)**.** *Under A1–A6, the expected per-side work (hash comparisons + level-entry handling, each $O(1)$) and the expected total* `uncertain` *entry count are*

$$W \;=\; \Theta\bigl(\min(b \cdot D,\, N) \cdot (1 + \log_b(N/D))\bigr)$$

- *for $D \le N/b$: $W = \Theta(b \cdot D \cdot (1 + \log_b(N/D)))$ — the prompt-level form $\Theta(D \cdot (1 + \log(N/D)))$ counts* disputed nodes visited*; the protocol pays a $\times b$ sibling fan on each (Lemma 3, §4);*
- *for $N/b < D \le N$ (heavily diverged / disjoint): the plateau vanishes and the peak term is $\Theta(N)$; $W = \Theta(N) = \Theta(D) \cdot O(b)$ — linear;*
- *additive $\Theta(D \cdot (1 + s_v))$ for absorbing/filtering the provided subtrees and the final collapse, and $\Theta(1)$ per round of fixed pipeline cost.* $\square$

Cross-checks: $D = 1$, $N = 10^6$ gives $W \approx 256 \cdot (1 + \log_b(N/b)) \approx 640$ vs. the §4 hand count of 527 ✓. Disjoint trees ($C = 0$, $N = D$) give $\Theta(N)$ ✓ (matches the one-shot character of bootstrap, §9.2).

Theorem 2 (bandwidth)**.** *Total bytes on the wire, both directions (each depth's* `uncertain` *is listed by exactly one side — the alternation splits levels between them):*

$$\begin{aligned} \textit{bytes} &= W \cdot (h + \bar{k}) && \textit{[}\texttt{uncertain}\textit{: 32-byte hash + k-byte prefix, } \bar{k} \le j_N + 1\textit{]} \\ &+ \Theta(D) \cdot (\bar{s} + s_v + {\le}32) && \textit{[}\texttt{providing}\textit{: payloads, versions, compressed spines]} \\ &+ O(D) \cdot \bar{k} && \textit{[}\texttt{requested}\textit{: one prefix per separated they-only subtree]} \\ &+ {\le}\, 36 \cdot 4 + 2 \cdot 25 + 2 \cdot s_v && \textit{[framing, preamble, handshake]} \end{aligned}$$

In Θ-terms: $\Theta(D \cdot (\bar{s} + s_v)) + \Theta(\min(b \cdot D, N) \cdot (1 + \log_b(N/D)) \cdot h)$. The empirical "$\sim D$" is right in its D-scaling; the per-D constant is the interesting part: at $N/D = 10^2 \ldots 10^4$ the hash channel runs $\sim$10–40 KB per differing message *(e.g. §4: 18 KB to move one message). For small payloads,* ***hash traffic dominates bandwidth****; bodies dominate only when $\bar{s} \gtrsim 36 \cdot W/D$ bytes.* □

Theorem 3 (latency)**.** *Messages strictly alternate and the conversation advances one depth per message (A5). Disputes empty at depth k^* (Lemma 2 corollary), plus $O(1)$ messages to flush the final* `requested` → `providing` *exchange; the in-band predicate then stops the session (A6 — no draining of the remaining schedule):*

$$\text{RTT} \;=\; \tfrac{1}{2} \cdot k^* + O(1) \;=\; \tfrac{1}{2} \cdot \log_{256}(2 \cdot D \cdot N) + c, \qquad c \approx 3\text{–}4$$

(preamble ≈ 1, handshake 1, `Initiate/Opening` *1, closing tail $\sim$1), hard-ceilinged at 36 frames $\approx$ 18 RTT, floored at $\sim$2 RTT total for equal versions. Path compression does not reduce this (A6). Numerically the* descent *portion is tiny at any scale — 1.8 levels $\approx$ 1 RTT for $(N, D) = (10^6, 1)$ up to 6.1 levels $\approx$ 3 RTT for $(10^7, 10^7)$; the $O(1)$ dominates. The empirical $\log_{256}(C + D)$ agrees at $D \approx N$ (where $\frac{1}{2}\log DN = \log N$) and is $\approx 2\times$ the true descent depth for $D \ll N$, a difference of < 2 RTT at any realistic scale — unfalsifiable from timing, trivially checkable by counting frames (§8).* □

7 Against the $\sqrt{C \cdot D}$ fit

Proposition 1 (the fit is a true upper bound)**.** *For $1 \le D \le C$:*

$$D \cdot \ln(C/D) \;\le\; \frac{2}{e} \cdot \sqrt{C \cdot D}, \qquad \textit{with equality iff } D = C/e^2.$$

Proof. Set $x = D/C \in (0, 1]$ and maximize $g(x) = \sqrt{x} \cdot \ln(1/x)$: $g'(x) = (\ln(1/x) - 2)/(2\sqrt{x}) = 0$ at $x = e^{-2}$, $g(e^{-2}) = 2/e$. Then $D \ln(C/D) = C \cdot x \ln(1/x) = C\sqrt{x} \cdot g(x) \cdots$ — directly, $D \cdot \ln(C/D) = \sqrt{CD} \cdot \sqrt{x} \cdot \ln(1/x) \le (2/e)\sqrt{CD}$. □

In base 256: $D \cdot \log_{256}(C/D) \le (2/(e \cdot \ln 256)) \cdot \sqrt{CD} \approx 0.133 \cdot \sqrt{CD}$. With Theorem 1's constants ($b/\ln b \approx 46.2$), for $D \le C$:

$$W \;\le\; b \cdot D \cdot (2 + \ln 2) + \frac{2}{e} \cdot \frac{b}{\ln b} \cdot \sqrt{CD} \;\approx\; 690 \cdot D + 34 \cdot \sqrt{C \cdot D} \;=\; O(\sqrt{C \cdot D}),$$

since $D \le \sqrt{CD}$ when $D \le C$. So the empirical $\sqrt{C \cdot D}$ is a *correct* upper bound on per-side computation — but loose: the ratio $\sqrt{CD}/(D \cdot \log(C/D)) = \sqrt{C/D}/\log(C/D)$ grows without bound as $D/C \to 0$, and the bound is tight (constant-factor) only in the band $D/C \in [\sim 10^{-2}, \sim\frac{1}{2}]$ where $\sqrt{x} \cdot \ln(1/x)$ stays within 1.5× of its max.

Why the fit nevertheless matched the data: the staircase. The smooth form $b \cdot D \cdot (1 + \log_b(N/D))$ is the envelope of a *staircase*: each level of the sum spans a factor of $256 \approx 2.4$ decades. Inside one span the dominant term is $t_k = D \cdot N/b^k$ — **linear in N at fixed D** — and the term above it, $\min(b, D) \cdot b^k$-shaped, is *flat* in D once $D > b$. The exact sum's local log-log slopes therefore wobble around the sqrt's 0.5 instead of sitting at the asymptotic 0 (in N) and 1 (in D):

Exact-sum values of W (children examined ≈ `uncertain` entries; computed from the master sum, expectations, constants ±2× from occupancy corrections), against $\sqrt{C \cdot D}$:

N	D	W	$\sqrt{C \cdot D}$	$W/\sqrt{CD}$	k^*	RTT (descent + c)
10^4	1	295	100	3.0	1.8	∼3.9
10^4	10^2	4.2 k	1.0 k	4.2	2.6	∼4.3
10^4	10^4	26 k	10 k	2.6	3.6	∼4.8
10^5	10^2	26 k	3.2 k	**8.2**	3.0	∼4.5
10^5	10^3	67 k	10 k	6.8	3.5	∼4.7
10^6	1	0.53 k	1.0 k	**0.53**	2.5	∼4.2
10^6	10^2	27 k	10 k	2.7	3.5	∼4.7
10^6	10^3	81 k	32 k	2.6	3.9	∼4.9
10^6	10^4	220 k	100 k	2.2	4.3	∼5.1
10^6	10^5	1.08 M	316 k	3.6	4.7	∼5.3
10^7	1	0.67 k	3.2 k	**0.21**	2.9	∼4.4
10^7	10^4	1.6 M	316 k	5.1	4.7	∼5.3
10^7	10^5	10.2 M	1.0 M	**10.2**	5.1	∼5.6
10^7	10^7	29 M	10 M	2.9	6.1	∼6.0

Read the ratio column: over five orders of magnitude in each axis it stays within $[0.2, 10]$ — a one-constant sqrt fit (say anchored at the central cells, $a \approx 2.6$) leaves residuals of only 0.1×–4×, and the worst residuals sit at the extreme corners ($D = 1$ at huge N; $D \approx b \cdot N/b^2$) where wall time is dominated by fixed per-session costs (preamble, handshake, ∼8 frames, thread/pipe synchronization in the bench harness) or by the $\Theta(D)$ body-handling term. Local log-log slopes of the exact sum, vs. sqrt's uniform 0.5:

- in D at fixed $N = 10^6$: 0.75, 0.96, **0.47, 0.43**, 0.69 per decade (D: $1 \to 10 \to 10^2 \to 10^3 \to 10^4 \to 10^5$) — *in the densest benchmark range the true slope is* ≈ 0.45*, indistinguishable from 0.5*;
- in N at fixed $D = 10^3$: 0.80, **0.08**, 0.43 per decade (N: $10^4 \to 10^5 \to 10^6 \to 10^7$) — the $10^5 \to 10^6$ decade is the smoking gun: the true cost is nearly flat there (it sits between staircase steps: $b^2 = 65{,}536$, $b^3 = 16.8\,\mathrm{M}$), while sqrt demands ×3.16.

Conclusion. $\sqrt{C \cdot D}$ and $\Theta(\min(bD, N)(1 + \log_b(N/D)))$ are empirically indistinguishable from wall time on diagonal or D-sweeps at grid scales; they separate cleanly on **fixed-D, N-sweeps across the $10^5 \to 10^6$ decade**, and on any *count*-based (rather than time-based) measurement.

8 An experiment that distinguishes them

All using existing infrastructure; no protocol changes.

1. **Byte counting (noise-free, decisive).** In `benches/gossip_grid.rs`, wrap the `PipeReader`/`PipeWriter` ends of `Wire` (`gossip_grid.rs:43-99`) in trivial `std::io::Read`/`Write` adapters that increment `AtomicU64`s, and record bytes per session per cell. Bandwidth is $\Theta(36 \cdot W + \text{body terms})$ by Theorem 2 and is exactly measurable. Regress $\log(\mathit{bytes} - a \cdot D)$ on $\log D$ and $\log(N/D)$ over cells with $D \geq 100$: the sqrt model predicts elasticity ≈ 0.5 in $\log(N/D)$; Theorem 1 predicts ≈ 0 (a log-of-log). Frame *counts* from the same adapter (count 4-byte headers) verify Theorem 3 against $\frac{1}{2} \cdot \log_{256}(2DN) + c$ to ± 1 frame.

2. **The flat decade.** Fix *differing* $= 1{,}000$, *redacted* $= 0$, sweep *common* $\in \{10^3, 10^4, 10^5\}$ (cells already exist; extend `COMMON` in `benches/support/grid.rs:74` to 10^6 for the decisive decade, accepting the 10-sample floor). Predicted wall-time ratios *common* $10^5 \to 10^6$: sqrt $\times 3.16$; Theorem 1 $\times 1.2$ (the 0.08-slope decade). One afternoon of Criterion time.

3. **Direct hash-compare counts.** Drive two `local::Exchange`s in-process through the `#[cfg(test)] mirror::mirror` entry (`mirror.rs:337-353`) with grid-shaped trees and count `uncertain` entries (e.g. instrument in a test, or just count from the byte totals: *entries* $\approx$ (*uncertain bytes*)$/(34 \ldots 64)$). This measures W itself, bypassing the $\alpha \cdot D + \mathit{fixed}$ wall-time floor entirely.

9 Caveats and edge regimes

1. **Lazy memoization (first session after writes).** A batch of m local changes leaves cold `OnceLock`s on the $\Theta(m \cdot (1 + \log_b(N/m)))$ ancestor nodes of the changed paths (`untyped.rs:40-47`; memos survive `Arc`-sharing, so *only* changed paths are cold, `untyped.rs:100-123`). The first `hash()` then pays one branch preimage per cold node — up to $1 + 33 \cdot 256 \approx 8.4\,\mathrm{KB}$ of BLAKE3 for a saturated branch, but only 34 bytes per compressed-spine level (`untyped.rs:283-287`; both conventions measured in `benches/hash_recon.rs`). This cost is additive, amortizes across sessions, and is excluded from the benches by `warm_caches()` (`gossip_grid.rs:154-160`).

2. **Bootstrap / one side empty ($D = N$).** No prefix has content on both sides, so there are *zero* disputes: the empty responder sends an empty `Opening` (`local.rs:337-341`), and the initiator's root-level drain Left-provides every root child in one message (`partition.rs:276-294`), filtered through `Unknown` against the empty version (everything survives). Cost: $\Theta(N)$ transfer and absorption, $O(1)$ rounds. Theorem 1's formula degrades gracefully to $\Theta(N)$ here, but the *mechanism* is different — the dispute machinery never engages.

3. **Deletion honoring.** A redacted leaf (present on A, deleted on B) keeps hashes differing along its path, so it costs descent exactly like a differing leaf — **count redactions in** D (the grid's throughput accounting already does: `grid.rs:115-127`). But it costs no body bytes: at separation, A's Left-provide is killed by the $O(1)$ ceiling check (`unknown.rs:28-34`) and A deletes its own copy (`partition.rs:232-239`) — deletion *propagates* through the version ceiling, with no tombstone and no transfer. A whole missing-but-dominated subtree dies the same way: one memoized comparison, no descent below it.

4. **Skew.** A1 is enforced by content addressing, not hoped for: because the version is folded into the leaf path (`path.rs:32-44`), even a million-message single-party version-batch spreads uniformly — concentration under few prefixes would require engineered BLAKE3 outputs. (Contrast key-ordered Merkle trees, where a contiguous batch *does* cluster.) Even granting an adversary clustered paths, the depth ceiling caps the damage at $W \leq 32 \cdot b \cdot D$ and 18 RTT.

5. **Version sizes.** ITC `Version`s are not constant-size: they ride the handshake, every provided leaf, and every `ceiling/floor` $\leq$ comparison (`unknown.rs:31, 46-49`), contributing the s_v factors in Theorems 1–2. For long-lived universes with many party splits, s_v is the term to watch, orthogonal to the tree combinatorics.

6. **The radix constant is the design tradeoff.** Per disputed node the protocol ships a full sibling fan (up to 256×36ish bytes $\approx 9\,\text{KB}$) to buy a 2.4-decade depth reduction per level. $b = 256$ puts every realistic deployment within $\sim$3 levels of separation (RTT-optimal) at the cost of the $\times b$ in W — which is exactly the constant that made a $\sqrt{C \cdot D}$ fit land in the right numeric range in the first place.